

PHYS 2210 MIDTERM 2 Exam - Resubmission

Due November 13, 2025 at 11:59pm

Instructor: C. De Grandi

In accordance with the University Student Code of Conduct:

- You are allowed to use: any material on our PHYS 2210 Fall' 25 Canvas site (including the formula sheet, lecture notes, homework solutions. etc.), our textbook, your class notes and formula sheet.
- You can communicate only with your classmates, only students currently enrolled in our class, in Discussion Section on Thursday November 13 and/or at other times you plan to meet with your study-group.
- You are not allowed to use a calculator.
- You are not allowed to access the internet.
- Communication, sharing, and discussion of exam content with anyone outside of your classmates in this class is prohibited.

Please sign with your first and last name in the box below to agree with the following statement:

I agree to adhere to these guidelines in accordance with the University Student Code of Conduct.

I understand that violating the Code of Conduct requirements will be considered Academic Misconduct.

If you are submitting as a group include the signatures of each group member.

Handwritten signatures and student IDs inside a box:

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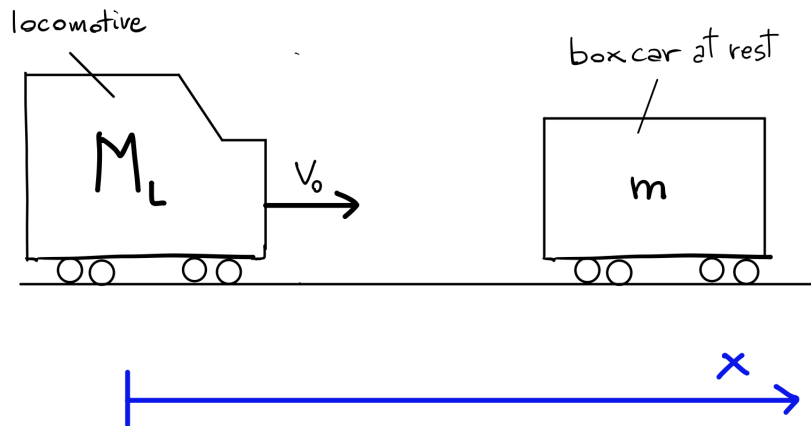
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Failure to comply with these guidelines will result in a determination of
Academic Misconduct in accordance with the Student Code
(Policy 6-400, Section V)

Problem 1: A locomotive and two boxcars (11 points)

A locomotive is approaching a train depot when its brakes fail. The depot team has designed a way to slow it down. The locomotive of mass M_L is made to collide with a boxcar of mass m at rest. The locomotive, initially moving with a speed v_0 , collides and couples with the boxcar. The collision can therefore be considered completely **inelastic** and the motion occurs along one direction, along an x-axis as shown in the figure below. Assume friction is negligible.



a) (4 points) Calculate v_1 , the speed of the locomotive+boxcar system right after the locomotive couples with the boxcar.

- Show clearly all the steps of your calculation and reasoning.
- Give the speed in terms of the given quantities only (M_L , m , v_0 , g).
- Simplify your answer as much as possible.
- Write your final answer inside the box shown below.

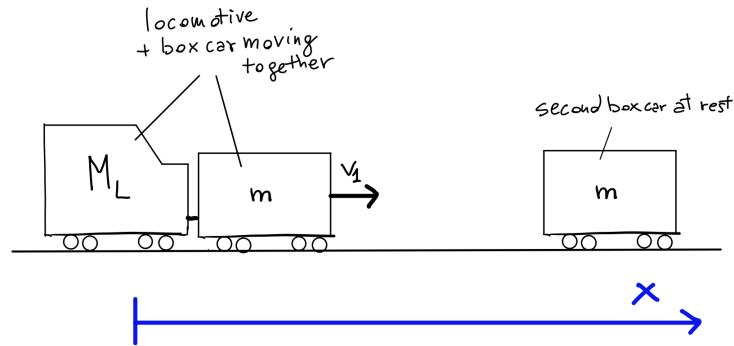
Solution: $M_L v_0 = (M_L + m) v_1$

$$v_1 = \frac{M_L v_0}{M_L + m}$$

Final Answer

$$v_1 = \frac{M_L v_0}{M_L + m}$$

Right after the collision the locomotive+boxcar system collides with a second boxcar of mass m at rest. The collision is, again, completely **inelastic**; the locomotive+boxcar couples with the second boxcar after the collision.



b) (4 points) Calculate v_2 , the speed of the locomotive+2 boxcars system right after the second collision.

- Show clearly all the steps of your calculation and reasoning.
- Give the speed in terms of the given quantities only (M_L , m , v_0 , g). Note v_1 is not a given quantity, and should not be in your final answer.
- Simplify your answer as much as possible.
- Write your final answer inside the box shown below.

Solution: $(M_L + m) V_1 = (M_L + 2m) V_2$

$$V_1 = \frac{M_L V_0}{M_L + m}$$

$$V_2 = \frac{(M_L + m) V_1}{M_L + 2m}$$

$$= \frac{M_L V_0}{M_L + 2m}$$

Final Answer

$$v_2 = \frac{M_L}{M_L + 2m} V_0$$

c) (3 points) The depot team measures that the speed of the locomotive after the two collisions is reduced to $2/3$ of its initial velocity, v_0 , before any collision. Calculate the mass of the locomotive in terms of m . Show the steps of your calculation. Simplify your answer.

Solution: $\frac{M_L}{M_L + 2m} V_0 = \frac{2}{3} V_0$

$$3M_L = 2M_L + 4m$$

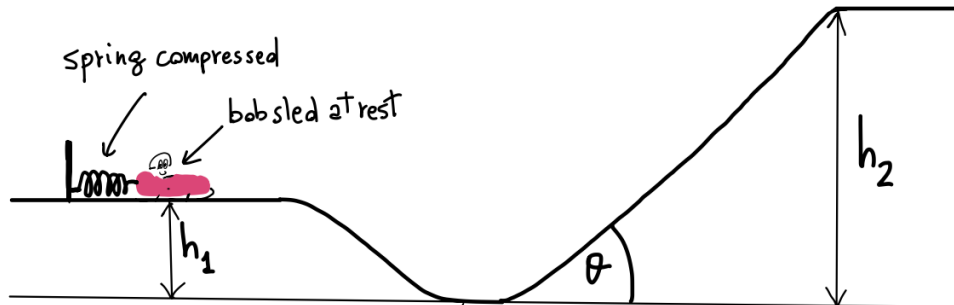
$$M_L = 4m$$

Final Answer

$$M_L = 4m$$

Problem 2: Bobsled and carpet (20 points)

A bobsled track made of ice is composed of: a horizontal surface at a height h_1 above ground, a downward ramp followed by an upward incline at an angle θ with the horizontal, and a horizontal surface at a height h_2 above ground (with $h_2 > h_1$) (as shown in the figure below). Everything is made of ice so the track is *frictionless*.



A big spring is attached on the left side of the first horizontal surface such that a bobsled of mass m can be set up to compress the spring and then be launched forward from rest. The spring constant is k .

a) (7 points) What is the *minimum* length $d_{\min}$ the spring needs to be compressed so that the bobsled will be able to arrive at the top of the incline, at a height h_2 ?

- Use the Generalized Work-Energy Theorem to find your answer and show the steps of your calculation.
- Give your answers in terms of the given quantities only: m , g , k , h_1 , h_2 , θ .
- Simplify your final answer as much as possible and write it inside the box shown below.
- *Hint: use dimensional analysis to check that your answer has units of length.*

Solution: $\frac{1}{2}kd^2 + mgh_1 = mgh_2$

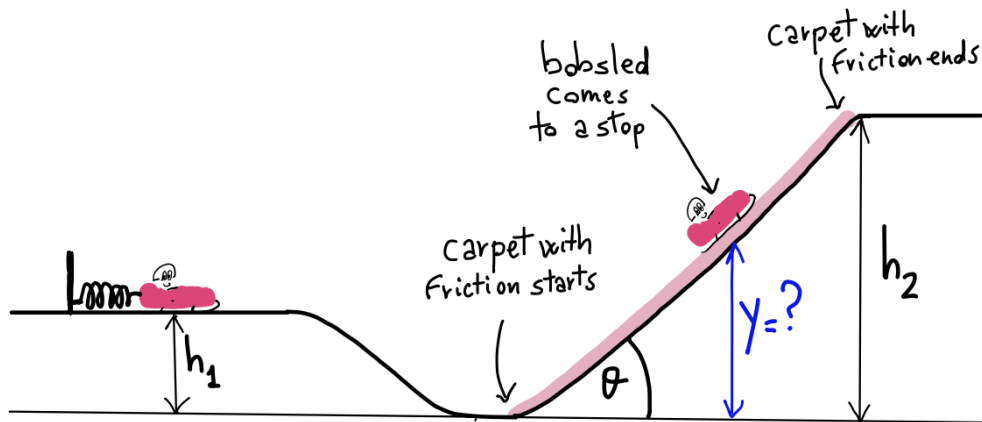
$$\frac{1}{2}kd^2 = mg(h_2 - h_1)$$

$$d_{\min} = \sqrt{\frac{2mg(h_2 - h_1)}{k}}$$

Final Answer

$$d_{\min} = \sqrt{\frac{2mg(h_2 - h_1)}{k}}$$

- b) (8 points) A carpet is laid above the entire length of the incline to slow down and bring the bobsled to a stop. Effectively the carpet acts as a **surface with friction** from the beginning of the incline (at height 0 above ground) to the top of the incline (at height h_2).



Assume:

- the coefficients of static and kinetic friction of the bobsled on the carpet are μ_s and μ_k , respectively;
- the bobsled starts with the spring compressed of the amount you found in the previous part;
- y is the height above ground at which the bobsled will come to a stop (as shown in the figure above)

Using the Generalized Work-Energy Theorem, find an expression for y in terms of the given quantities **only**: m , g , k , h_1 , h_2 , θ , μ_s , μ_k . Show the steps of your calculation, simplify your final answer as much as possible and write it in the box shown below.

Solution: $E_i = \frac{1}{2} k d_{\min}^2 + mgh_1 = mgh_2$

$$W_f = - f_k x = - \mu_k mg \cos \theta \cdot \frac{y}{\sin \theta}$$

$$= - \mu_k mgy \cot \theta$$

For $E_i + W_f = mgy$

we solve:

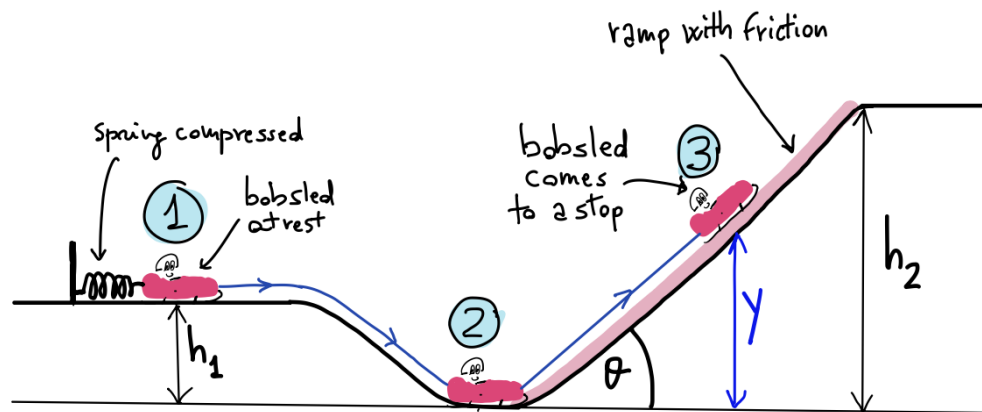
$$mgh_2 - \mu_k mgy \cot \theta = mgy$$

$$y = \frac{h_2}{1 + \mu_k \cot \theta}$$

Final Answer

$$y = \frac{h_2}{1 + \mu_k \cot \theta}$$

- c) (5 points) The figure below summarizes the entire motion of the bobsled from the previous part: 1) the bobsled is at rest and the spring is compressed, 2) the bobsled is at the bottom of the ramp, 3) the bobsled has come to a stop along the incline at the height y that you determined in the previous part.



Call K_1 , K_2 , and K_3 the kinetic energy of the bobsled at instant 1), 2), and 3) respectively (as marked in the figure above).

Call U_1 , U_2 , and U_3 the potential energy of the bobsled (including both spring and gravity potential) at instant 1), 2), and 3) respectively (as marked in the figure above).

Using appropriately the signs: $>$, $<$, $=$, rank the kinetic energies K_1 , K_2 , and K_3 , from biggest to smallest.

Using appropriately the signs: $>$, $<$, $=$, rank the potential energies U_1 , U_2 , and U_3 , from biggest to smallest.

No explanation required.

Your ranking for K :
(Biggest $\rightarrow$ Smallest)

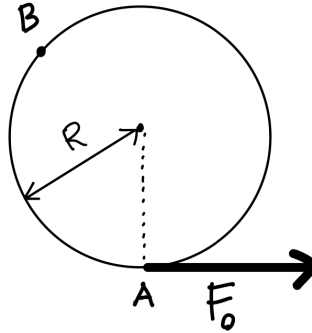
$$K_2 > K_1 = K_3$$

Your ranking for U :
(Biggest $\rightarrow$ Smallest)

$$U_1 > U_3 > U_2$$

Problem 3: Torque on disk (8 points)

The figure below shows a top view of a solid disk of mass M and radius R . A force F_0 is applied at the point A on the circumference of the disk as shown in the figure. The force acts along a direction tangential to the circumference.



a) (2 points) What is the magnitude of the torque caused by the force F_0 around the center of the disk? Circle your answer.

A) $\tau = 2RF_0$

B) $\tau = R^2F_0$

☒ C) $\tau = RF_0$

D) $\tau = \frac{R}{2}F_0$

b) (2 points) What is the direction of the torque caused by the force F_0 around the center of the disk? Circle your answer.

A) Into the page

☒ B) Out of the page

C) Upwards

D) Downwards

c) (4 points) Assume the disk is initially at rest and that the force applied causes a *constant* angular acceleration α . Calculate the **tangential speed** v of the point B on the circumference of the disk after a time t_1 . Give your answer in terms of α , t_1 and the given quantities (R , M , F_0). Show the steps of your calculation, simplify your final answer as much as possible and write it in the box shown below.

Solution: $\omega = \alpha t_1$

$$v = R\omega = R\alpha t_1$$

$$\tau = RF_0, I = \frac{1}{2}MR^2$$

$$\alpha = \frac{\tau}{I} = \frac{2F_0}{MR}$$

$$v = \frac{2F_0}{M} t_1$$

Final Answer

$$v = \frac{2F_0}{M} t_1$$

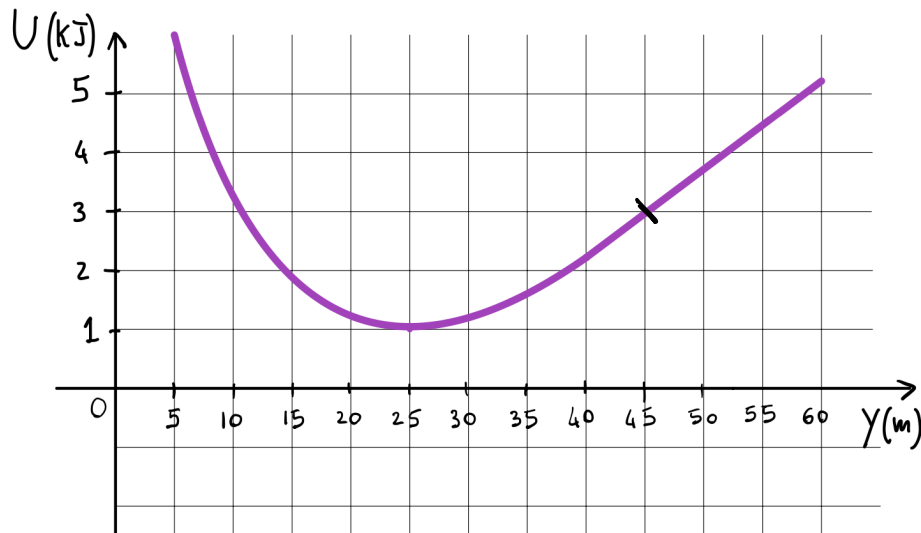
Problem 4: Energy Diagram (11 points)

Bungee jumping is an activity which involves jumping from a tall structure (e.g. a bridge) while connected to an elastic cord.

- a) (7 points) The figure below shows a plot of the total potential energy (gravitational and elastic) for the jumper as a function of the height y from the ground (note the units on the graph are meters (m) for the horizontal axis and kiloJoules (kJ) on the vertical axis). Consider a fearful bungee jumper who (too scared of jumping) just lets himself fall from the bridge with zero initial velocity. His maximum kinetic energy during the fall is measured to be 2 kiloJoules.

For this entire problem neglect air resistance and any form of friction.

- By looking at the graph, extract: 1) the height of the bridge from where the jumper started his fall, 2) the height above the ground where the jumper will stop for a moment before being pulled back up by the bungee cord.
- **Show and explain the steps and reasoning you used to find your answers.**
- Give your answers in meters and write them inside the box shown below. Make sure to include the units with your answers.



$$E = K_{\max} + U_{\min} = 2 + 1 = 3 \text{ kJ}$$

$$3 \text{ kJ} \Rightarrow \begin{array}{ll} 45 \text{ m} & \text{bridge} \\ 5 \text{ m} & \text{stop} \end{array}$$

Final Answers

$$h_{\text{bridge}} = 45 \text{ m} \quad h_{\text{stop}} = 5 \text{ m}$$

- b) (4 points) Consider the total net force acting on the jumper when he reaches the maximum kinetic energy. Is the net force: zero, upwards or downwards? Circle your answer.

☒ A) Zero

☐ B) Upwards

☐ C) Downwards

Give an explanation for your answer in words and/or in equations in the space below.

Solution: When net force is zero,

it speed $\uparrow$ before it

it speed $\downarrow$ after it

so it is the max speed position